

IMPROVING PROACTIVE AI ASSISTANCE WITH HIERARCHICAL PROCEDURAL UNDERSTANDING

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ABSTRACT

Proactive AI assistants continuously observe a user’s activity and decide whether to provide new guidance or remain silent. To effectively assist users, they should provide appropriate guidance for the task, determine when to provide the next guidance based on the current task progress, and adjust the guidance level to the user’s expertise and needs. To realize these capabilities, proactive AI assistants require appropriate training and evaluation data. However, existing datasets either remain limited to detection-based proactive understanding or provide procedural guidance at a fixed granularity. Fixed-granularity guidance alone provides limited information about what has been completed and what remains, making it difficult to track progress and determine completion in a streaming setting. It also provides limited context about the purpose and expected outcome of the current event and limits the ability to adapt guidance granularity to the user’s needs. To address these limitations, we introduce the *ProactiveCoach* suite, comprising *ProactiveCoach-Instruct* for training, *ProactiveCoachBench* for evaluation, and fine-tuned vision-language models (VLMs) with an adaptive guidance system. *ProactiveCoach-Instruct* provides hierarchically structured guidance at the phase, step, and action levels, offering richer supervision for learning task progress and procedural context across different granularities. *ProactiveCoachBench* evaluates whether models provide appropriate guidance at the right time across different guidance levels and whether they can adapt when the requested guidance level changes. We fine-tune pretrained VLMs on the *ProactiveCoach-Instruct* dataset and demonstrate its effectiveness across different backbones. Compared with fixed-granularity supervision, hierarchical supervision consistently improves overall performance across various backbones, with gains of up to 9.6%p. We further build an adaptive guidance system by combining our fine-tuned model with a lightweight guidance router. Without additional fine-tuning of the guide model, our system outperforms the in-context adaptation baseline by 57.1%p in overall performance across four guidance-level transitions. Our project page is available at <https://jinsuby.github.io/ProactiveCoach/>.

1 INTRODUCTION

Building AI assistants like JARVIS¹ that continuously understand their surroundings and proactively help users when needed has long been a human aspiration. For example, a future smart-glasses assistant could recognize what a user is currently doing and provide guidance at an appropriate moment, and a future humanoid assistant robot could monitor task progress and proactively provide physical assistance to the user when help is needed (Wang et al., 2026; Xia et al., 2026; Zheng et al., 2026; Zhao et al., 2026; Azad et al., 2026; Zhang et al., 2026; Yan et al., 2026a; Ma et al., 2026; Yu et al., 2026). A central requirement for realizing such AI assistants is *proactiveness*.

¹JARVIS stands for “Just A Rather Very Intelligent System.” It is a fictional AI assistant used by Tony Stark in the Marvel Cinematic Universe, first appearing in *Iron Man* (2008). It assists Stark by managing information and controlling various systems and devices.

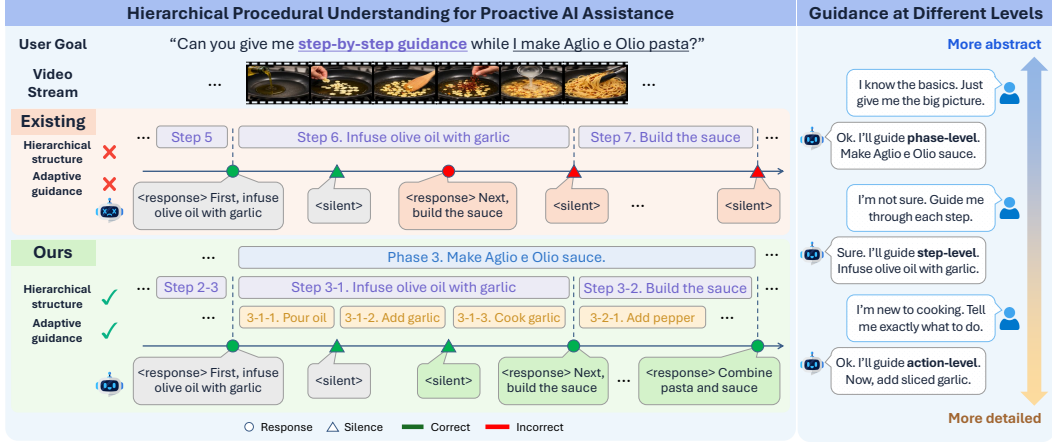


Figure 1: Overview of PROACTIVECOACH. It tracks progress and completion across phase, step, and action levels and adapts the guidance level to the user’s request.

Proactiveness refers to continuously observing the user’s activity in a streaming setting and deciding whether to provide new guidance or remain silent. Building on this notion of *proactiveness*, a proactive AI assistant requires three core capabilities. (1) *It should provide appropriate guidance for the user’s task.* Beyond simply recognizing objects, the assistant should understand the current situation and task progress in activities such as cooking, assembly, and repair, which require a sequence of events to achieve a goal. (2) *It should provide the next guidance after the currently guided event is completed.* This requires tracking the progress of the guided event and providing the next guidance when it ends, while remaining silent when no new guidance is needed. (3) *It should provide guidance at a level appropriate to the user’s expertise and needs.* The assistant should adjust the guidance granularity accordingly and adapt when the requested guidance level changes during an ongoing task.

To build proactive AI assistants with the capabilities defined above, appropriate data are essential for training and evaluation. Although recent studies have introduced training datasets and benchmarks for proactive assistance, existing data do not sufficiently support these core capabilities. Some datasets remain limited to detection-based proactive understanding, where models detect the occurrence of specific objects or events in a video stream and respond accordingly (Li et al., 2025; Lin et al., 2026; Wang et al., 2025; Zhao et al., 2026; Li et al., 2026b). Other datasets provide procedural guidance for the user’s task, but mainly provide guidance at a fixed granularity (Zhang et al., 2025b; Li et al., 2026a; Kundu et al., 2026; Liu et al., 2026). Since a guided event may consist of multiple finer-grained events, fixed-granularity guidance provides limited information about what has been completed and what remains, making it difficult to track progress and determine completion in a streaming setting. Since a guided event also belongs to a broader procedure, such guidance provides limited context about its purpose and expected outcome, making it difficult to understand how the current event contributes to the overall procedure. Fixed-granularity guidance further limits the ability to adjust the guidance level to the user’s expertise and needs or adapt when the requested level changes.

To address these limitations, we introduce the *ProactiveCoach* suite for proactive AI assistance, comprising a training dataset, an evaluation benchmark, and fine-tuned VLMs. Our main contributions are as follows. (1) We introduce *ProactiveCoach-Instruct* for training and *ProactiveCoach-Bench* for evaluation. ProactiveCoach-Instruct provides hierarchically structured guidance at the phase, step, and action levels, together with guidance timing, enabling models to learn both fine-grained task progress and higher-level procedural context and to generate guidance at different granularities. ProactiveCoachBench evaluates whether models provide appropriate guidance at the right time at each guidance level, and further assesses whether they can adapt when the requested guidance level changes during an ongoing task. (2) We fine-tune pretrained VLMs on ProactiveCoach-Instruct to jointly learn phase-, step-, and action-level proactive guidance in a single model. This enables different VLM backbones to learn hierarchical procedural understanding and determine what guidance to provide and when to provide it at each level. (3) We build an adaptive guidance system by combining the fine-tuned VLM with a lightweight guidance router. The VLM generates multi-level

Dataset	#Videos	Hours	Proactive	Procedural Tasks	Guidance Tasks	Hierarchical Ann.
HiREST	3.4K	0.3K	✗	✓	✗	✓
GUIDE	3.5K	0.1K	✗	✓	✗	✓
Ego4D-HCap	4.5K	–	✗	✗	✗	✓
Action100M	1.2M	128K	✗	✓	✗	✓
ProAssist	3.4K	0.5K	✓	✓	✓	✗
StreamPro-Bench	0.6K	–	✓	△	△	✗
Pro ² Bench	29.3K	–	✓	✓	✓	✗
GuideMe	2.5K	0.2K	✓	✓	✓	✗
ProactiveCoach (Ours)	7.2K	1.5K	✓	✓	✓	✓

Table 1: Comparison of existing datasets for hierarchical video understanding and proactive procedural assistance. △ denotes partial task coverage.

guidance, while the router selects the guidance level requested by the user, allowing the system to dynamically adjust guidance granularity during an ongoing task without additional fine-tuning of the guide model.

Our proposed dataset is constructed using Procedural Data Curation, Procedure Hierarchy Construction, and Proactive Guidance Generation, achieving an overall score of 4.27 in our human quality review. We also demonstrate that our training data can be effectively applied across different pre-trained VLM backbones. Across all four backbones, hierarchical supervision enables multi-level guidance while improving step-level Avg. by 6.3–9.6%p over step-only supervision. Finally, our adaptive guidance system enables proactive assistance to adapt its guidance granularity based on the user’s needs. Without additional fine-tuning of the guide model, it outperforms the in-context adaptation baseline by 57.1%p in Avg. across four guidance-level transitions.

2 RELATED WORK

Hierarchical Video Understanding. Hierarchical video understanding aims to understand video content at multiple levels, from fine-grained actions to higher-level activities (Song et al., 2023; Islam et al., 2024; Kang et al., 2025). HiREST (Zala et al., 2023) connects video and moment retrieval with step-by-step descriptions, while GUIDE (Liang et al., 2024) identifies common procedures across multiple videos and links them to the detailed steps in each video. Ego4D-HCap (Islam et al., 2024) recursively generates captions at multiple levels, from short action descriptions to long-form video summaries, while Action100M (Chen et al., 2026) hierarchically segments videos and provides action and context annotations for segments of different durations. However, hierarchical descriptions cannot be directly converted into guidance. In proactive procedural assistance, the assistant must identify goal-relevant activities and determine what guidance to provide, when to provide it, and at what level of detail.

Proactive Procedural Assistance. Unlike reactive AI, which typically responds to explicit user queries, proactive AI must determine not only *what to say* but also whether and *when to respond* based on a continuous stream of observations. Some datasets remain limited to detection-based proactive understanding, where models detect specific objects or events in a video stream and respond accordingly (Li et al., 2025; Lin et al., 2026; Wang et al., 2025; Zhao et al., 2026; Li et al., 2026b). More recently, proactive procedural assistance has moved beyond such detection-based settings, requiring models to determine what guidance is needed and when to provide it based on the user’s task goal and ongoing task context (Zhang et al., 2025b; Li et al., 2026a; Kundu et al., 2026; Liu et al., 2026).

Proactive procedural assistance aims to provide task guidance, requiring models to learn what guidance to provide and when to provide it. ProAssist (Zhang et al., 2025b) and StreamPro-Bench (Li et al., 2026a) focus on providing proactive step-level guidance according to task progress. Pro²Bench (Kundu et al., 2026) further considers recognizing when the user’s behavior does not follow the intended procedure and providing recovery guidance, while GuideMe (Liu et al., 2026) covers task completion verification, error detection, and corrective guidance. However, existing datasets and benchmarks primarily focus on step-level guidance, providing limited supervision and evaluation for understanding progress and completion across procedural levels. They also provide limited



Figure 2: Overview of hierarchically structured guidance in PROACTIVECOACHBENCH.

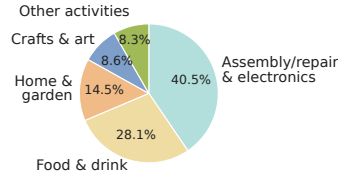


Figure 3: Domain distribution of our dataset.

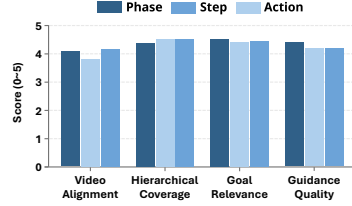


Figure 4: The quality of PROACTIVECOACHBENCH.

evaluation of whether models can provide guidance across different procedural levels and adapt when the requested guidance level changes.

3 PROACTIVECOACH-INSTRUCT & PROACTIVECOACHBENCH

In this section, we introduce ProactiveCoach-Instruct, our training dataset, and ProactiveCoach-Bench, our evaluation benchmark. We describe their design and statistics in Sec. 3.1, present the construction pipeline in Sec. 3.2, and show quality review in Sec. 3.3

3.1 DATASET OVERVIEW

To support hierarchical procedural understanding for proactive procedural assistance, we structure goal-relevant guidance into three levels: phase, step, and action. This hierarchy allows lower-level progress to provide evidence for the completion of guided events, while higher-level context helps clarify their purpose and expected outcome. Based on this design, we construct PROACTIVECOACH-INSTRUCT and PROACTIVECOACHBENCH with hierarchically structured guidance and guidance timing at each level, supporting learning and evaluation across procedural levels.

Figure 2 illustrates the hierarchically structured guidance in ProactiveCoach-Instruct and ProactiveCoach-Bench. A phase is a high-level unit that achieves an intermediate goal through multiple steps, while a step is a procedural unit that achieves a specific outcome through multiple actions. An action is an observable fine-grained event that can be guided independently and serves as the finest guidance level. Each sample consists of a task goal, a video, and hierarchical guidance annotations, with each guidance sentence linked to its event span and guidance time. Guidance timing is defined independently at each level. The next guidance is provided immediately after the previously guided event at that level ends. For each video chunk, we combine multi-level guidance in phase, step, and action order into a single **<response>**. If no new guidance is required at any level, the chunk is labeled **<silent>**.

The data contain 11,119 samples constructed from 7,232 first-person videos across eight source datasets: Ego4D Goal-Step (Song et al., 2023), Ego-Exo4D (Grauman et al., 2024), EPIC-KITCHENS (Damen et al., 2018), CaptainCook4D (Peddi et al., 2024), EgoExoLearn (Huang et al.,

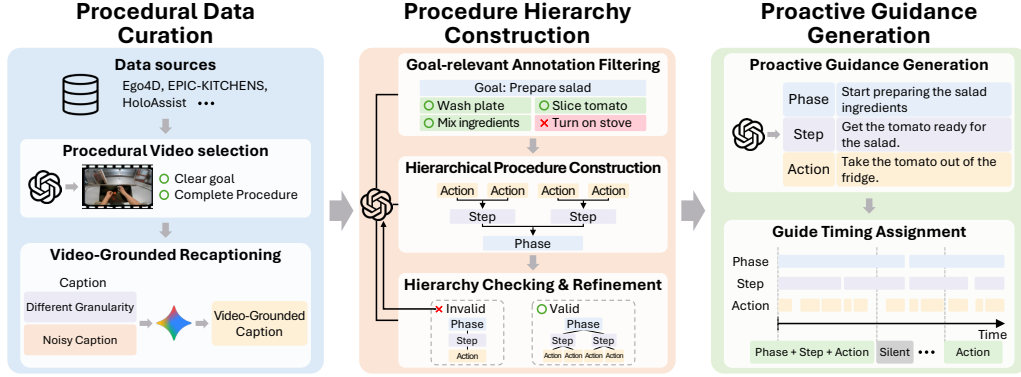


Figure 5: Construction pipeline for our proposed dataset.

2024), HD-EPIC (Perrett et al., 2025), WTaG (Bao et al., 2023), and HoloAssist (Wang et al., 2023). As shown in Figure 3, the data cover diverse domains, including assembly, repair, and electronics; food and beverage preparation; and household and gardening activities. We use 10,007 samples as ProactiveCoach-Instruct for training and 1,112 samples as ProactiveCoachBench for evaluation. On average, each sequence contains 4.14 phases, with 3.35 steps per phase and 2.47 actions per step. Further dataset statistics are provided in Appendix A.

3.2 CONSTRUCTION PIPELINE

The proposed dataset is constructed through three stages: procedural data curation, procedure hierarchy construction, and proactive guidance generation, as shown in Figure 5.

Procedural Data Curation. We filter goal-oriented procedural videos from eight egocentric video datasets. The source datasets may contain videos that are unsuitable for procedural assistance, such as those with unclear task goals or insufficient coverage of key events. Therefore, in procedural video selection, we use GPT-4.1-mini to assess the clarity of the task goal and the coverage of key events based on the existing annotations, and retain videos suitable for procedural assistance. In addition, the source datasets contain annotations with different levels of granularity, inconsistencies with the video, or missing relevant events. To address these issues, we perform video-grounded recaptioning using Gemini 2.5 Flash with the selected videos and their existing annotations. The model revises captions that do not match the video, adds missing captions, and aligns each caption with its corresponding temporal segment in the video.

Procedure Hierarchy Construction. The recaptioned annotations may still include content that is not necessary for accomplishing the task goal. Therefore, in goal-relevant annotation filtering, we use GPT-4.1-mini to retain only the annotations necessary for achieving the task goal. Then, we perform hierarchical procedure construction to organize the selected annotations into a procedural hierarchy. First, we consolidate the selected annotations into actions, each representing an independent unit of execution. We then group multiple actions into a step representing a procedural unit, and multiple steps into a phase representing an intermediate goal, forming a phase-step-action hierarchy. The temporal span of each action is defined by its observed start and end times in the video. The temporal spans of each step and phase are then defined to cover their child actions and steps, respectively. This produces a hierarchical procedure structure that serves as the basis for hierarchically structured guidance. However, this process may produce redundant structures where different hierarchical levels describe the same execution. We therefore perform hierarchy checking and refinement to eliminate these redundancies and ensure hierarchical consistency. Using GPT-4.1-mini, we check whether each step is properly composed of its actions, each phase is properly composed of its steps, and their temporal spans are consistent. Based on this check, we reconstruct the hierarchy and repeat the refinement process until the structure is consistent.

Proactive Guidance Generation. To construct hierarchically structured guidance, we perform level-specific guidance generation. Given the constructed phase-step-action hierarchy and the user’s goal information at each level, we generate proactive guidance for the phase, step, and action levels by using GPT-4.1-mini. Then, we perform guide timing assignment to align the generated guidance

with task progress based on the temporal spans at each level. At each level, the next guidance is assigned to the frame immediately after the current guided event ends. If the current guided event ends at only one level, we provide the next guidance only for that level. If guided events end at multiple levels at the same time, we provide the next guidance for all corresponding levels together. If no guided event ends at a given frame, we label that frame as silent. Through this process, we construct *ProactiveCoach-Instruct* and *ProactiveCoachBench* with hierarchically structured guidance for proactive AI assistance.

3.3 QUALITY REVIEW

We conduct a human evaluation to assess whether the annotations accurately reflect the procedural structure of the video and provide appropriate guidance. We sample 50 examples from *ProactiveCoachBench* and evaluate them at the phase, step, and action levels using four criteria: *video alignment*, *hierarchical coverage*, *goal relevance*, and *guidance quality*. *Video alignment* measures whether the annotated content, temporal spans, and guidance timings are consistent with the observed video. *Hierarchical coverage* evaluates whether the lower-level executions sufficiently cover the content required by the higher-level procedure, while *goal relevance* measures whether the generated guidance contributes to achieving the task goal. *Guidance quality* evaluates whether each guidance sentence clearly and accurately describes what the user should do at the corresponding level. As shown in Figure 4, the annotations achieve scores above 4.0 on most evaluation criteria, with an overall score of 4.27. These results indicate that the annotations reliably capture video-grounded procedural hierarchies and provide goal-relevant guidance at each level. Details of quality review are provided in Appendix D.

4 PROACTIVECOACH

We propose ProactiveCoach, a hierarchical guidance learning method that jointly learns phase-, step-, and action-level proactive guidance in a single vision-language model (VLM). Using *ProactiveCoach-Instruct*, we train the model to learn hierarchical procedural understanding across the three levels. For supervised fine-tuning (SFT), we organize guidance across the three levels into a single output sequence that preserves their hierarchical relationships and guidance timing. Each guidance sentence is paired with its level and a hierarchical ID that encodes the parent-child structure. This makes the relationship between lower-level events and higher-level procedural units explicit in the training targets.

Let I denote the instruction specifying the task goal, V_t^w the video frames from up to w consecutive chunks ending at the current chunk t , and y_t the ground-truth output for that chunk. We denote the ground-truth output history by $H_t^{\text{GT}} = (y_1, \dots, y_{t-1})$. Given I , V_t^w , and H_t^{GT} , the model is trained to output `<silent>` when no new guidance is needed and `<response>` followed by the corresponding guidance otherwise. If new guidance is needed at one level, the response contains guidance for that level. If new guidance is needed at multiple levels, the corresponding guidance is combined into a single response. During training, we use teacher forcing and provide previous ground-truth outputs as the output history.

In procedural videos, `<silent>` labels are more frequent than `<response>` labels. To prevent repeated silence from dominating the loss, we focus supervision on transitions between response and silence, following a prior work (Li et al., 2026c). Specifically, we select all `<response>` chunks and each `<silent>` chunk immediately following a `<response>` chunk for supervision. We then randomly sample additional `<silent>` chunks from the remaining chunks, matching the total number of chunks selected above. We compute the loss only on output tokens selected for supervision. The objective is as follows:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\frac{1}{M} \sum_{t=1}^T m_t \log p_{\theta}(y_t \mid I, V_t^w, H_t^{\text{GT}}). \quad (1)$$

Here, T is the total number of chunks, and $m_t \in 0, 1$ indicates whether chunk t is selected for supervision. We normalize the loss by $M = \sum_{t=1}^T m_t$, the total number of selected chunks. Training on hierarchically structured guidance encourages the model to understand how actions make up each step and how each step belongs to a higher-level phase. Understanding how each action is progressing can provide useful evidence for determining whether the current guided step is complete,

while higher-level phase context can help clarify the purpose of the current step and its expected outcome. Together with the observed video, this hierarchical understanding provides a stronger basis for deciding what guidance to provide and when to provide it at each level.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Datasets and Baselines. We evaluate on ProactiveCoachBench and the publicly available validation split of EgoProactive, an existing benchmark for proactive procedural assistance. We evaluate guidance performance at the phase, step, and action levels on ProactiveCoachBench, and at the step level on EgoProactive (Kundu et al., 2026). We compare against four groups of models: Proprietary Frontier Models (Comanici et al., 2025; Anthropic, 2025; Singh et al., 2025), Multimodal Models (Bai et al., 2025; Team, 2026; Yan et al., 2026b; Zhu et al., 2025; Zhang et al., 2025a; 2024), Guidance Assistant Models (Zhang et al., 2025b), and Streaming Models (Chen et al., 2024; Wang et al., 2026; Li et al., 2026c). For guidance assistant models, we use ProAssist, whose model weights are publicly available, and for streaming models, we use VideoLLM-Online, MMDuet2, and VideoChat3-4B as baselines. The details of baselines are provided in Appendix E.2.

Evaluation Protocol. Following prior evaluation protocols (Kundu et al., 2026; Liu et al., 2026), we evaluate models at predefined anchor points. These anchor points are categorized into *Response Anchors* and *Silent Anchors*. At *Response Anchors*, the model is expected to generate appropriate guidance, whereas at *Silent Anchors*, it is expected to remain silent and avoid providing unnecessary guidance. For ProactiveCoachBench, we use all ground-truth guidance timestamps as *Response Anchors*. For *Silent Anchors*, we randomly sample a timestamp with no ground-truth guidance from the central 30–70% of each phase, step, or action segment. For EgoProactive, we use the original evaluation anchor points. At each anchor, all models receive the most recent 10 seconds of video preceding the anchor, together with the complete ground-truth dialogue history prior to that point. We use ground-truth dialogue history rather than model-generated responses to prevent previous incorrect or missed guidance from affecting subsequent decisions, enabling a fair comparison across models under the same dialogue history. This protocol therefore evaluates both whether the model correctly decides when to provide guidance and whether the provided guidance is appropriate.

Metrics. We use sF1, gF1, and PQS to evaluate proactive procedural assistance, following prior work (Kundu et al., 2026; Liu et al., 2026).

sF1 jointly evaluates how well the generated guidance matches the ground-truth guidance and whether the model provides all necessary guidance. We first match ground-truth and generated guidance one-to-one based on both timing and content, and sum the semantic similarities of the matched pairs. Then, we compute sPrecision and sRecall by dividing this sum by the numbers of generated and ground-truth guidance instances, respectively. sF1 is calculated as the harmonic mean of sPrecision and sRecall (Liu et al., 2026).

gF1 evaluates whether the model correctly decides when to provide guidance and when to remain silent. It considers only whether the model responds or remains silent at each anchor point and does not consider the guidance content. We compute precision and recall separately for response and silence, obtain the corresponding F1 scores, and calculate gF1 as their geometric mean (Kundu et al., 2026).

PQS jointly evaluates response–silence decisions and the quality of the provided guidance. At a silent anchor, the model receives a score if it remains silent and receives zero otherwise. At a response anchor, the model receives a score based on the relevance, specificity, actionability, and conciseness of the generated guidance, and receives zero if it remains silent. PQS is computed by averaging these scores across all anchor points (Kundu et al., 2026). We use GPT-5.2 to evaluate guidance content quality. Details of the metrics are provided in Appendix F.

Implementation Details. We use Qwen3-VL-4B/8B and Qwen3.5-4B/9B as backbone models and train for three epochs using eight B200 GPUs. Videos are sampled at 1 FPS and grouped into 2-second chunks, with the model deciding whether to respond at each chunk. We apply a causal block-sparse attention mask that restricts each token to frames from the current and preceding 10 seconds,

Model	Ours	ProactiveCoachBench												EgoProactive			
		Phase				Step				Action							
		sF1	gF1	PQS	Avg.↑	sF1	gF1	PQS	Avg.↑	sF1	gF1	PQS	Avg.↑	sF1	gF1	PQS	Avg.↑
Proprietary Frontier Models																	
Gemini 3 Flash Preview		43.01	70.36	52.87	55.41	41.15	59.00	41.88	47.34	45.08	57.45	36.88	46.47	38.62	64.69	50.73	51.35
Claude Haiku 4.5		43.25	27.70	18.18	29.71	41.51	22.54	22.89	28.98	41.60	23.90	19.80	28.43	38.81	52.63	34.20	41.88
GPT-5.4 mini		33.66	50.03	31.23	38.31	32.47	51.21	34.47	39.38	34.47	53.87	35.82	41.39	39.71	53.60	36.18	43.17
Guidance Assist Models																	
ProAssist-8B		11.06	38.86	49.59	33.17	9.45	33.72	49.03	30.73	9.83	30.96	48.45	29.75	6.07	24.64	45.74	25.49
Streaming Models																	
VideoLLM-Online		0.45	8.97	49.97	19.80	0.32	6.48	49.95	18.92	0.34	5.99	49.98	18.77	0.52	9.44	46.17	18.71
MMDuet2		9.97	40.41	41.98	30.79	19.96	48.25	37.93	35.38	19.80	49.25	36.38	35.14	30.63	44.71	27.53	34.29
VideoChat3-4B		35.91	19.12	14.38	23.14	35.12	25.63	19.85	26.87	37.83	18.00	16.64	24.16	39.46	36.32	24.21	33.33
Multimodal Models																	
VideoChat-R1.5-7B		0.05	2.41	49.81	17.42	0.07	3.93	49.72	17.91	0.11	3.33	49.43	17.62	0.20	4.67	45.69	16.85
InternVL3-8B		13.86	45.11	41.13	33.37	25.12	52.14	38.96	38.74	34.71	54.90	34.36	41.32	18.04	38.08	43.38	33.17
VideoLLaMA3-7B		15.90	44.46	35.63	31.99	21.66	48.82	38.15	36.21	27.74	50.08	34.16	37.33	1.07	9.14	41.80	17.34
LLaVA-Video-7B		20.34	47.74	35.26	34.45	30.19	51.65	34.90	38.92	25.34	51.55	40.10	39.00	35.73	47.65	33.05	38.81
Qwen3-VL-32B		41.56	38.50	22.11	34.06	38.46	23.89	21.71	28.02	38.85	20.84	17.25	25.65	32.06	49.81	40.14	40.67
Qwen3-VL-4B		35.93	25.82	18.19	26.65	36.50	26.01	20.99	27.83	37.46	27.78	18.65	27.96	39.93	35.90	24.33	33.39
	✓	53.15	74.40	60.53	62.69	49.30	71.58	55.99	58.96	49.72	64.69	45.32	53.24	36.48	62.78	46.50	48.59
Qwen3-VL-8B		40.93	9.54	17.35	22.60	38.39	9.56	20.03	22.66	37.70	8.99	15.82	20.84	41.96	2.47	23.06	22.50
	✓	54.63	76.21	61.37	64.07	50.24	72.02	55.83	59.37	50.07	64.12	44.77	52.99	35.28	63.06	48.55	48.96
Qwen3.5-4B		19.20	47.85	35.71	34.25	20.86	49.10	39.90	36.62	21.76	47.81	40.81	36.79	14.84	36.68	44.22	31.91
	✓	52.87	75.15	60.79	62.94	49.35	72.27	56.27	59.30	50.43	64.75	45.15	53.44	35.76	60.06	45.61	47.14
Qwen3.5-9B		40.10	11.65	17.42	23.06	38.38	26.90	21.54	28.94	40.71	26.86	18.92	28.83	37.10	25.85	21.87	28.27
	✓	50.50	72.79	60.08	61.13	48.24	71.29	56.90	58.81	50.56	66.17	47.58	54.77	36.67	62.17	46.73	48.52

Table 2: Proactive guidance results on ProactiveCoachBench and the EgoProactive validation set.

while preserving access to the ground-truth instruction and all previous response and silence tokens. Detailed training configurations are provided in Appendix E.1.

5.2 MAIN RESULTS

As shown in Table 2, our method consistently achieves higher overall performance than the base-lines across all guidance levels on ProactiveCoachBench. A higher sF1 indicates that the generated guidance is semantically well aligned with the ground-truth guidance while sufficiently covering the task goal. A higher gF1 indicates that the model more accurately determines when to provide guidance and when to remain silent. A higher PQS indicates better overall performance in both response-silence decisions and guidance quality. Our method shows strong performance across all metrics, with consistent improvements across different backbones.

Our method also shows strong performance across all metrics on the EgoProactive validation set. Some base models achieve relatively high sF1 but very low gF1, indicating that they tend to respond even when no guidance is needed. Conversely, some models show relatively low sF1 but high PQS, indicating that they tend to remain silent even when a response is required. Our method consistently achieves higher Avg. scores than the base models across various backbones.

5.3 ABLATION STUDIES AND ANALYSIS

Hierarchical vs. Step-Only Supervision. Table 3 compares step-only supervision with hierarchical supervision that jointly learns phase-, step-, and action-level guidance across different backbones. Hierarchical supervision consistently improves sF1, gF1, PQS, and Avg. across all backbones. These results show that learning hierarchically structured guidance not only enables guidance at multiple levels but also improves step-level proactive assistance.

Effect of Phase- and Action-Level Supervision. Table 4 compares step-only, phase-step, step-action, and phase-step-action supervision to analyze the effect of additional guidance levels. Adding either phase- or action-level supervision improves step-level performance over step-only supervision, and using all three levels achieves the highest performances. These results suggest that action-level supervision helps the model understand what actions make up each step and how they are

Method	ProactiveCoachBench Step			
	sF1	gF1	PQS	Avg.
<i>Qwen3-VL-4B</i>				
Step-only	42.85	63.34	41.78	49.32
Multi-level (Ours)	49.30	71.58	55.99	58.96
<i>Qwen3-VL-8B</i>				
Step-only	45.82	66.12	47.27	53.07
Multi-level (Ours)	50.24	72.02	55.83	59.37
<i>Qwen3.5-4B</i>				
Step-only	41.01	64.53	44.62	50.06
Multi-level (Ours)	49.35	72.27	56.27	59.30
<i>Qwen3.5-9B</i>				
Step-only	42.56	65.21	45.77	51.18
Multi-level (Ours)	48.24	71.29	56.90	58.81

Table 3: Step-only vs. hierarchical supervision.

Method	ProactiveCoachBench Step			
	sF1	gF1	PQS	Avg.
<i>Qwen3.5-4B</i>				
Step-only	41.01	64.53	44.62	50.06
Phase-Step	48.21	66.22	46.37	53.60
Step-Action	46.16	70.81	55.02	57.33
Phase-Step-Action	49.35	72.27	56.27	59.30

Table 4: Ablation on supervision levels.

FT	Phase				Step				Action			
	sF1	gF1	PQS	Avg.	sF1	gF1	PQS	Avg.	sF1	gF1	PQS	Avg.
✓	8.53	32.52	46.92	29.32	7.40	29.23	44.33	26.99	38.44	11.10	15.67	21.74
✓	52.87	75.15	60.79	62.94	49.35	72.27	56.27	59.30	50.43	64.75	45.15	53.44

Table 5: Zero-shot multi-level guidance.

progressing, providing useful evidence for determining whether the step is complete. Phase-level supervision provides complementary context about the purpose and expected outcome of the current step and how it contributes to the overall procedure. These results support the benefit of learning hierarchically structured guidance for step-level proactive assistance.

Zero-Shot Multi-Level Guidance. We examine whether a base model can perform multi-level proactive guidance through prompting alone, without fine-tuning on hierarchically structured guidance. Table 5 compares zero-shot Qwen3.5-4B with our fine-tuned model under the same multi-level prompt and three-level GT history. Fine-tuning substantially improves performance across all three guidance levels. These results suggest that prompting alone is insufficient for reliable multi-level proactive guidance, while training on hierarchically structured guidance helps the model learn the relationships across procedural levels and provide appropriate guidance at each level.

Guidance-Level Adaptation. A practical proactive procedural assistant should adapt its guidance level when the user’s request changes during an ongoing task. As illustrated in Figure 6, our system separates guidance generation from guidance-level selection. Our trained model generates guidance at the three levels, while a lightweight LLM router selects which level to present based on the user’s request. Because the two modules operate asynchronously and independently, the displayed guidance level can change without modifying or fine-tuning the guide model.

We evaluate this on four level transitions: Step→Phase, Step→Action, Phase→Step, and Action→Step. For the baseline, the level-change request is added to the dialogue history, and the model is prompted to adjust its guidance level accordingly. In our system, the router handles the level change while the guide model continues to generate multi-level guidance. As shown in Table 6, our system substantially improves performance across all transition scenarios. For the baseline, the previous dialogue history contains guidance consistently provided at the original level, making it

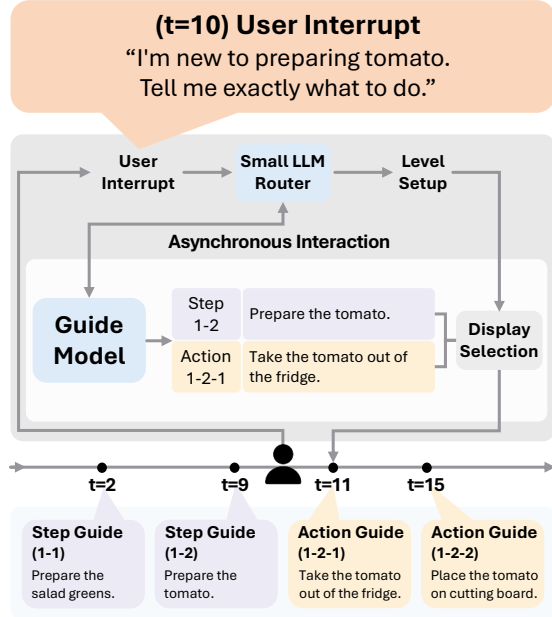


Figure 6: Guidance-level adaptation with a lightweight router. ProactiveCoach generates multi-level guidance, while the router selects the level requested by the user.

difficult to adapt to a new level through in-context prompting alone. In contrast, separating guidance generation from level selection allows our system to change the displayed guidance level without being constrained by the previous guidance history. These results show that our design can reliably adapt guidance granularity to the user’s request without additional fine-tuning of the guide model.

6 CONCLUSION

In this work, we explored hierarchical understanding for proactive procedural assistance. Existing datasets primarily focus on step-level guidance, providing limited supervision and evaluation across procedural levels and for guidance-level adaptation. We introduced ProactiveCoach-Instruct for training and ProactiveCoachBench for evaluation, providing hierarchically structured guidance at the phase, step, and action levels with guidance timing. We further proposed ProactiveCoach, a hierarchical guidance learning method that jointly learns this multi-level guidance in a single vision-language model. Our method consistently improves proactive assistance across different backbones and supports guidance-level adaptation during an ongoing task. We believe that our proposed dataset and method will provide a useful foundation for future research on proactive procedural assistance.

AI USE STATEMENT

In this work, we used generative AI tools for literature exploration, research and experimental design, code development, manuscript drafting, translation and editing, figure preparation, dataset construction, and automated evaluation. We have reviewed all AI-assisted work and take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

Our proposed dataset is constructed from publicly available video datasets and does not involve the collection of new personal information. We use the source datasets in accordance with their respective licenses and terms of use, and any released data will follow the same conditions.

REPRODUCIBILITY STATEMENT

To support reproducibility, we provide details of the dataset construction pipeline, training and evaluation settings, prompts, and metrics in the main paper and appendix. The supplementary materials include the training and inference code, evaluation implementation, and additional details for reproducing our experiments. We also provide an anonymous repository with a dataset preview and related materials, and plan to release the data and code.

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Appendices

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A MORE DETAILS OF PROACTIVECOACH-INSTRUCT AND PROACTIVECOACHBENCH

Our dataset contains 11,119 samples constructed from 7,232 first-person videos across eight source datasets. As shown in Table 7, 10,007 samples are used for ProactiveCoach-Instruct and 1,112 samples for ProactiveCoachBench. Samples from the same video are assigned to the same split. As shown in Figure 7, the average sample duration is 7.8 minutes and the median is 8.8 minutes, with durations ranging from 40 seconds to 18 minutes. The total duration is 1,452.1 hours. Each sample includes the corresponding events, level-specific guidance, and guidance times. The data contain 582,084 guidance targets in total: 46,074 at the phase level, 154,507 at the step level, and 381,503 at the action level. On average, each sample contains 4.14 phases, with 3.35 steps per phase and 2.47 actions per step.

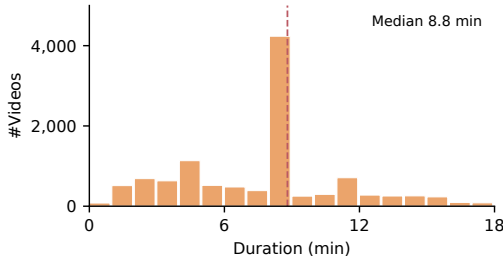


Figure 7: Sample duration distribution.

Property	Train	Test	Total
Samples	10,007	1,112	11,119
Unique video files	6,517	715	7,232
Duration (hours)	1,306.9	145.1	1,452.1

Table 7: Dataset split statistics.

B DATASET CONSTRUCTION DETAILS

We construct ProactiveCoach-Instruct and ProactiveCoachBench following the pipeline described in Sec. 3.2 and Figure 5: procedural data curation, procedure hierarchy construction, and proactive guidance generation. We provide prompt details for each stage. We use Gemini 2.5 Flash for video-grounded recaptioning, GPT-4.1-mini for the remaining generation and filtering steps, and GPT-4.1 for hierarchy checking and refinement. Guide timing assignment follows a rule-based procedure.

B.1 PROCEDURAL DATA CURATION

Figures 8 and 9 present the prompts for procedural video selection and video-grounded recaptioning, respectively.

Procedural Video Selection (Stage 1-1)

System.

You select videos for goal-directed procedural assistance.

Input.

`video_metadata` and `source_captions`.

Requirements.

1. Assess whether the ordered captions describe a clear task goal and a recognizable outcome.
2. Score goal clarity, coverage of key operations, and logical order from 1 to 10. Repeated motions do not count as distinct procedural stages.
3. Return an overall suitability score and a short evidence-based reason. Do not assume missing operations occurred.
4. Based on these scores and reasoning, decide whether the video should be selected for procedural assistance.

Output.

A JSON object with `goal_guess`, `goal_score`, `coverage`, `order`, `overall_score`, `reasoning`, and `selected`.

Figure 8: Prompt for Procedural Video Selection.

Video-grounded Recaptioning (Stage 1-2)**System.**

You produce video-grounded captions for an egocentric procedural video.

Input.

`video_clip`, `clip_metadata`, and `source_captions`.

Requirements.

1. Use source captions as references and verify each caption against the video. Preserve supported content, correct inaccurate descriptions, supplement visible omissions, and reject unsupported captions.
2. Make the caption granularity consistent across source datasets. Each caption should describe a single observable fine-grained manipulation.
3. Align each caption's event interval with the corresponding event observed in the video.
4. Describe only actions, objects, and state changes that are supported by the video. Do not infer unobserved actions or states. Use English descriptions.

Output.

A JSON object with `captions`. Each caption contains `id`, `description`, `object`, `state_change`, `event_interval`, and its source caption reference. `object` and `state_change` may be null when they are not clearly observable in the video. Newly added captions may have no source caption reference.

Figure 9: Prompt for Video-grounded Recaptioning.

B.2 PROCEDURE HIERARCHY CONSTRUCTION

Figures 10, 11, and 12 present the prompts for goal-relevant annotation filtering, hierarchical procedure construction, and hierarchy checking and refinement, respectively.

Goal-Relevant Annotation Filtering (Stage 2-1)**System.**

You select annotations that contribute to the supplied task goal. Output only JSON.

Input.

`task_goal` and `video-grounded_annotatons`.

Requirements.

1. Judge each annotation in the context of the task goal and surrounding annotations. Keep annotations that directly contribute to the task or support a necessary operation.
2. Retain preparation and supporting events, such as fetching an ingredient, measuring, or opening a container, when they serve the goal. Repeated events are not automatically irrelevant.
3. Exclude activity unrelated to the task goal, such as an unrelated conversation or camera adjustment. Do not reject an annotation merely because it does not yet fit a Step.
4. Preserve annotation IDs, descriptions, objects, state changes, and event intervals. Give an evidence-based reason for exclusions; do not invent missing operations.

Output.

A JSON object with `retained_ids` and `excluded`; each `excluded` entry contains `annotation_id` and `reason`. Account for every input annotation exactly once.

Figure 10: Prompt for Goal-Relevant Annotation Filtering.

Hierarchical Procedure Construction (Stage 2-2)**System.**

You organize observed activity into a Phase–Step–Action hierarchy.

Input.

`task_goal`, `retained_annotations`, and `review_feedback` (if any).

Requirements.

1. Consolidate the retained annotations into observable Actions that are meaningful to guide independently. Use their descriptions, objects, state changes, and event intervals to determine which annotations belong to the same Action. Merge annotations that describe the same event while preserving distinct events, and set each Action interval to cover the event intervals of its annotations.
2. Group Actions that together form a procedural task into a Step, and group Steps that together achieve an intermediate goal into a Phase. Define levels by procedural scope, not sentence length.
3. Set each Step interval to cover its Action intervals, and each Phase interval to cover its Step intervals.
4. Each Action belongs to one Step and each Step to one Phase. Descriptions at a higher level must express the collective scope of the lower-level units they contain. Avoid a Phase–Step–Action chain that merely renames the same operation; do not create additional Actions or Steps only to increase branching.
5. When review feedback is supplied, reconstruct the affected groups using the retained annotations and preserve valid assignments. Return explicit Phase–Step, Step–Action, and annotation references.

Output.

A JSON object with `phases`, `steps`, and `actions`. Each unit contains `id`, `description`, `event_interval`, and `annotation_ids`. Each Step contains its `phase_id`, and each Action contains its `step_id`.

Figure 11: Prompt for Hierarchical Procedure Construction.

Hierarchy Checking and Refinement (Stage 2-3)**System.**

You review the scope and evidence support of a proposed procedural hierarchy.

Input.

`task_goal`, `retained_annotations`, and `candidate_hierarchy`.

Requirements.

1. Check whether each lower-level unit contributes to the higher-level unit that contains it. Use the retained annotation descriptions, objects, state changes, and event intervals as evidence, and check whether relevant observed operations are missing or duplicated.
2. Flag higher-level units whose scope is not supported by the lower-level units they contain, as well as Phase–Step–Action chains that express the same operation across levels. A lower-level unit count alone is not sufficient; inspect the actual procedural scope.
3. Check Phase–Step and Step–Action references, chronological order, and whether each higher-level event interval covers the event intervals of the lower-level units it contains. Separate temporal errors from semantic grouping errors.
4. Return evidence-based reconstruction feedback for affected units to 2-2. Preserve valid groups and do not invent unobserved operations.

Output.

A JSON object with `passed`, `issues`, and `reconstruction_feedback`. Each issue identifies the affected unit, issue type, annotation IDs, and requested correction.

Figure 12: Prompt for Hierarchy Checking and Refinement.

B.3 PROACTIVE GUIDANCE GENERATION

Figure 13 presents the prompt for level-specific guidance generation. In Guide Timing Assignment (Stage 3-2), guidance timing is determined independently at the Phase, Step, and Action levels. Within the same level, the next guidance is assigned to the first frame after the current event ends. The first guidance at each level is assigned to the first frame of the video. When guidance from multiple levels is assigned to the same frame, we combine it into a single `<response>` in Phase, Step, and Action order. If no guidance is assigned at any level, the frame is labeled `<silent>`.

Level-Specific Guidance Generation (Stage 3-1)

System.

You write proactive instructions that match the scope of each Phase, Step, and Action.

Input.

`task_goal`, `hierarchy_units`, and `supporting_annotations`.

Requirements.

1. Use the task goal, hierarchy context, and supporting annotations to write an actionable imperative instruction in English.
2. Phase guidance conveys an intermediate goal; Step guidance conveys a procedural task; Action guidance conveys a concrete manipulation.
3. Do not describe a higher-level task using only its first lower-level operation. Do not include content from another unit or add unsupported details. Avoid filler.
4. Check each instruction against its corresponding Phase, Step, or Action and supporting annotations, correcting inaccurate objects, unsupported details, or scope mismatches. Preserve the supplied unit IDs. Generate content only; timing is assigned separately.

Output.

A JSON object with `guidance`; each entry contains `unit_id`, `level`, and `instruction`.

Figure 13: Prompt for Level-Specific Guidance Generation.

C EXAMPLES OF PROACTIVECOACHBENCH

These six examples illustrate the hierarchically structured guidance in ProactiveCoachBench. Red denotes the task goal, and blue shows phase-, step-, and action-level guidance at the first six consecutive guidance times.

TASK: clean and lubricate bicycle chain

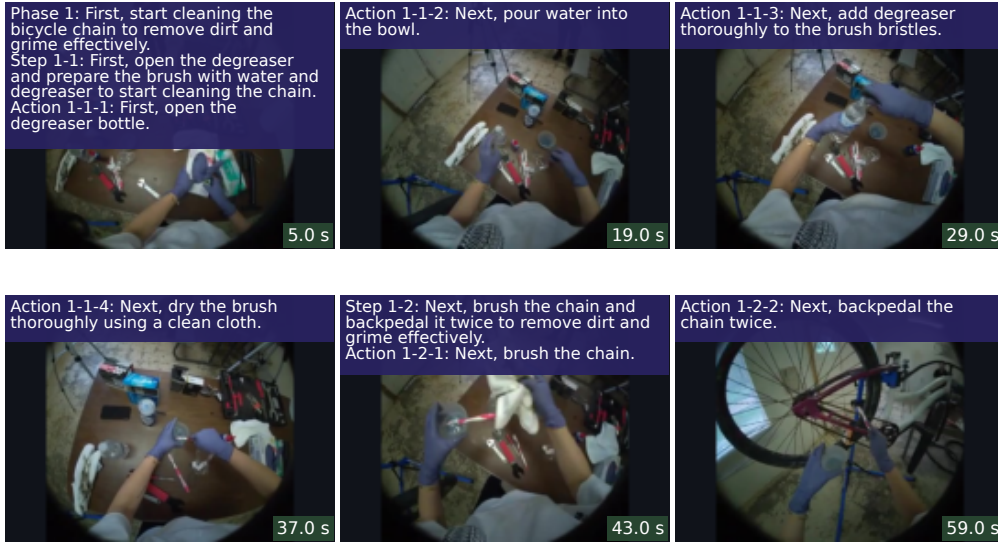


Figure 14: ProactiveCoachBench example-1 (goal: clean and lubricate a bicycle chain).

TASK: making a tossed vegetable salad

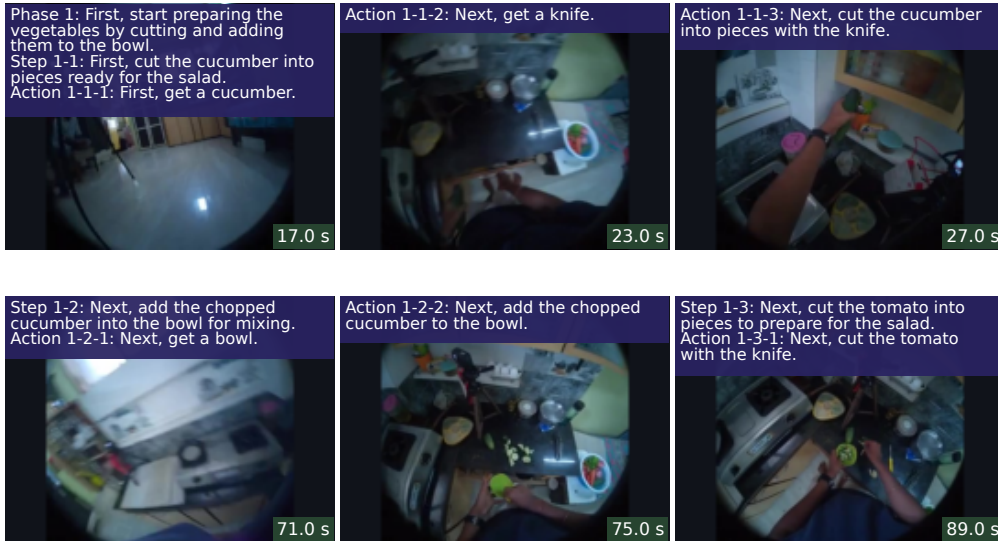


Figure 15: ProactiveCoachBench example-2 (goal: make a tossed vegetable salad).

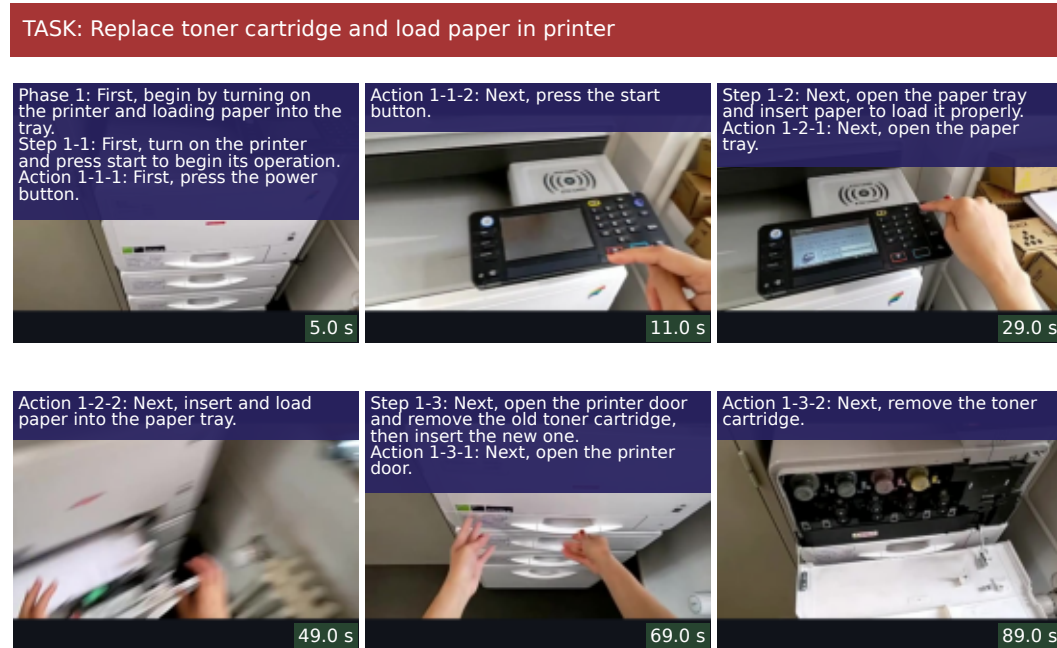


Figure 16: ProactiveCoachBench example-3 (goal: replace a toner cartridge and load paper).

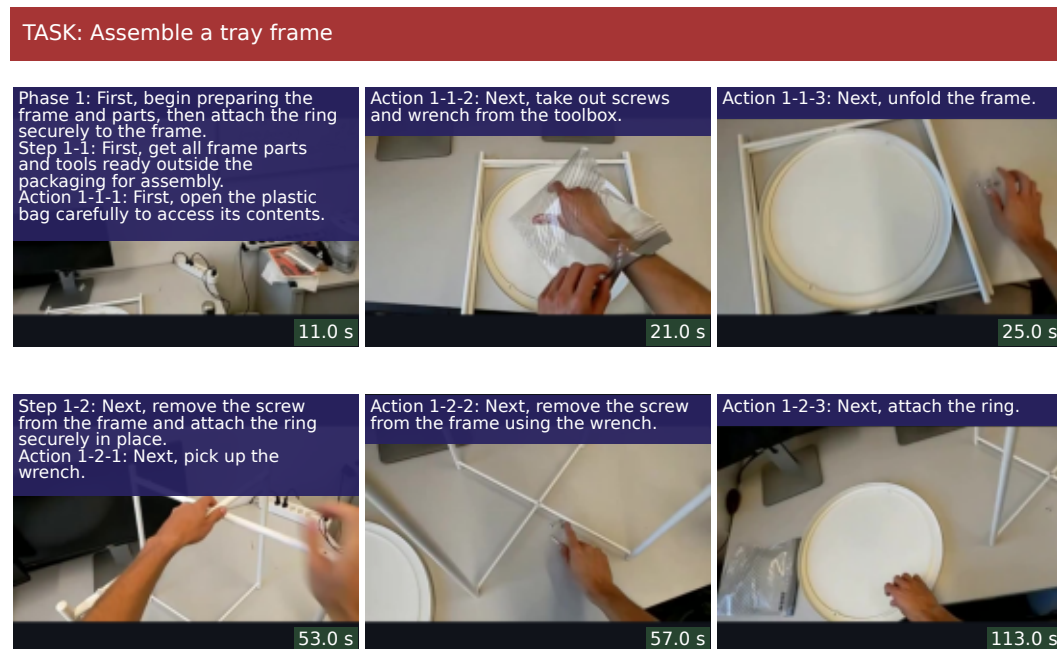


Figure 17: ProactiveCoachBench example-4 (goal: assemble a tray frame).

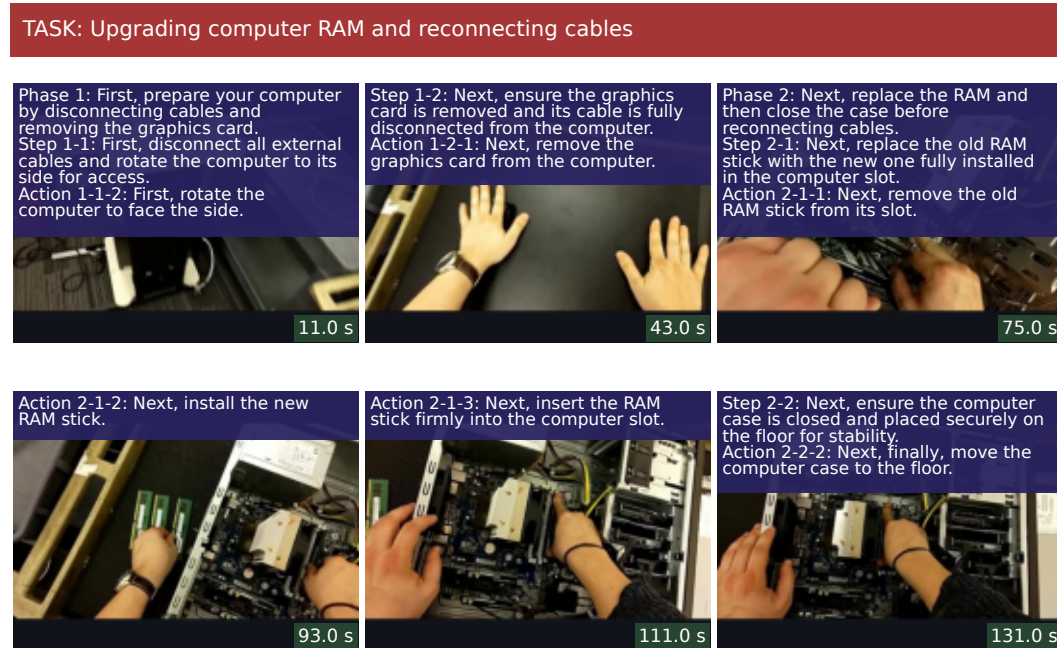


Figure 18: ProactiveCoachBench example-5 (goal: upgrade computer RAM and reconnect cables).

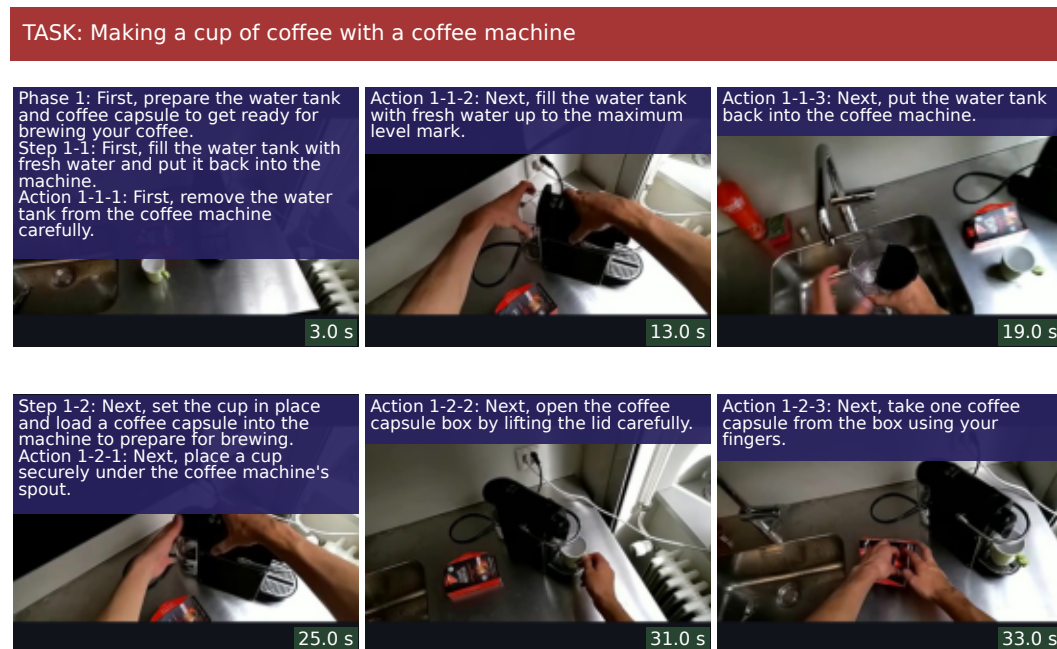


Figure 19: ProactiveCoachBench example-6 (goal: make a cup of coffee).

D DETAILS OF QUALITY REVIEW

We conduct a human quality review on 50 test examples at the phase, step, and action levels using four criteria: *video alignment*, *hierarchical coverage*, *goal relevance*, and *guidance quality*. Figure 20 shows the interface used for the evaluation.

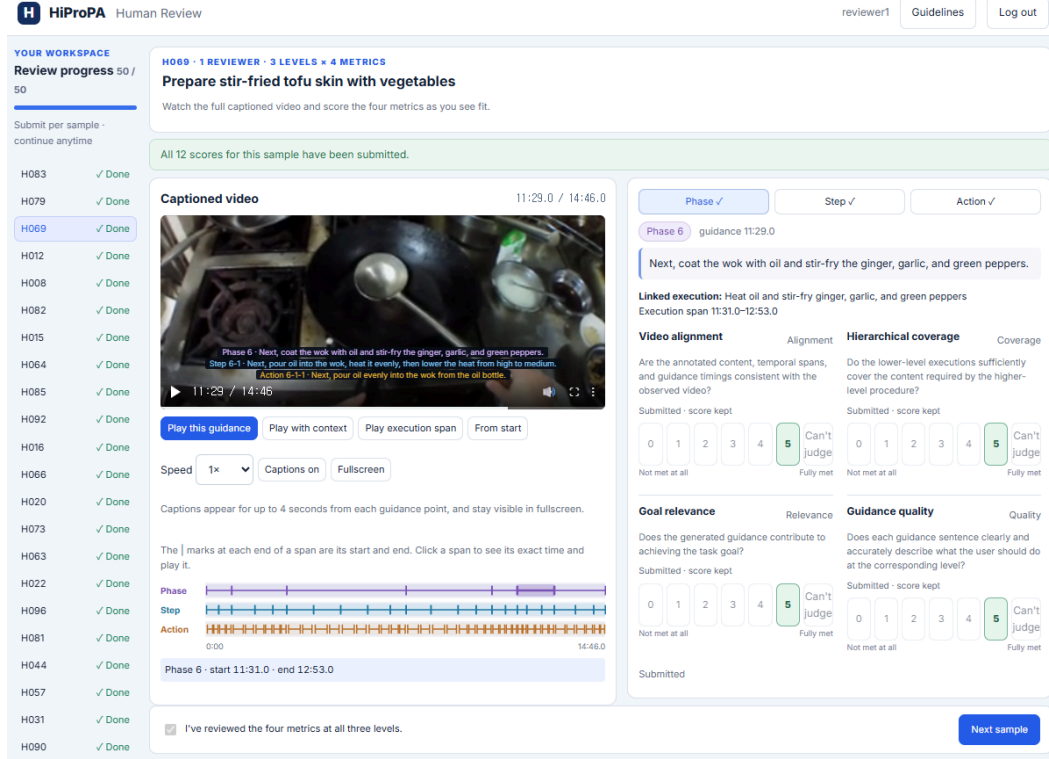


Figure 20: Human quality review interface. Reviewers inspect the task video and its hierarchical annotations at the phase, step, and action levels and evaluate them using four criteria: video alignment, hierarchical coverage, goal relevance, and guidance quality.

Video alignment evaluates whether the annotated content, temporal spans, and guidance timing are consistent with the observed video. Guidance may be provided before the next execution begins. However, guidance that is provided excessively early or late, or unnecessarily repeats a completed execution, is rated lower. The scores are 4.05, 4.14, and 3.80 for the phase, step, and action levels, respectively, showing that the annotations generally align well with the video progress.

Hierarchical coverage evaluates whether lower-level executions sufficiently cover the content required by the higher-level procedure. We examine whether the phases cover the activities needed to achieve the goal, the steps cover each phase, and the actions cover each step, focusing on whether any essential execution is missing. The scores are 4.35, 4.50, and 4.48, respectively, indicating strong coverage across all levels.

Goal relevance evaluates whether the selected executions and guidance contribute to achieving the task goal. Preparatory, safety-related, and recovery events that are necessary to achieve the task goal are considered relevant, whereas events unrelated to the goal are rated lower. The scores are 4.48, 4.41, and 4.39, respectively, showing high goal relevance across all levels.

Guidance quality evaluates whether each guidance sentence clearly and accurately describes what the user should do at the corresponding level. Lower scores are assigned when the guidance is unclear, difficult to follow, or does not match the intended level of detail. The scores are 4.37, 4.16, and 4.16, respectively, indicating that the generated guidance is generally clear and appropriate for each level.

E DETAILS OF EVALUATION PROTOCOL

E.1 TRAINING CONFIGURATIONS

Table 8 summarizes the training settings. We keep the original task instructions. Video chunks and target outputs are ordered by time, with each output placed at its reference timestamp. We used sliding-window causal attention, restricting direct access to video tokens from the latest five chunks (10 seconds), including the current chunk, while retaining task instructions and prior output text within the context limit.

Setting	Shared configuration
Training hardware	8 NVIDIA B200 GPUs
Optimizer	AdamW
Learning rate	2×10^{-5}
Learning-rate schedule	Cosine decay; 5% warmup
Weight decay	0.1
Gradient clipping norm	1.0
Training epochs	3
Effective batch size	32 sequences
Maximum sequence length	98,304 tokens
Numerical precision	BF16
Distributed training	DeepSpeed ZeRO stage 1
Video sampling rate	1 fps
Video chunk	2 frames

Table 8: Shared SFT configurations.

E.2 EVALUATION PROMPTS

Figure 21 shows the inputs shared by all baseline models: the task goal, the most recent 10 seconds of video, the ground-truth dialogue history, and the requested guidance level. Frontier models and multimodal baselines, including Qwen, use the shared prompt in Figure 22. Figure 23 shows the prompts for streaming and proactive assistance models, and Figure 24 shows the prompt for our model.

Common Inputs
Task Goal. {task_goal}
Requested Guidance Level. {Phase / Step / Action} Phase: a high-level unit that achieves an intermediate goal through multiple steps. Step: a procedural unit that achieves a specific outcome through multiple actions. Action: an observable fine-grained event that can be guided independently.
Ground-Truth Dialogue History. At {past_time}s: {past_GT_guidance} Additional ground-truth guidance strictly preceding the current anchor, if any. Past guidance describes what was said, not proof of completion.
Current Observation. Current anchor: {time}s. The most recent 10 seconds of video preceding the current anchor.

Figure 21: Common inputs for evaluation.

Frontier and Multimodal Baselines

You are a proactive assistant helping a user perform a procedural task. Use the current observation and ground-truth dialogue history to determine whether new guidance is needed at the requested guidance level. If guidance is needed, provide one concise instruction that helps achieve the task goal. Output one line: `$interrupt$` followed by the instruction, or `$silent$` when no new guidance is needed.

Based on the current observation and previous guidance, determine whether new guidance is needed at the requested guidance level. If needed, provide one concise instruction for the task.

Figure 22: Shared evaluation prompt for frontier models and multimodal baselines, including Qwen.

Streaming and Proactive Assistance Models**VideoLLM-Online and ProAssist.**

You are a proactive assistant helping a user perform a procedural task. Use the current observation and previous guidance to determine whether new guidance is needed at the requested guidance level. Use the native streaming gate to remain silent when no new guidance is needed. When the gate selects a response, provide one concise instruction that helps achieve the task goal.

Based on the current observation and previous guidance, determine whether new guidance is needed at the requested guidance level. If needed, provide one concise instruction for the task.

VideoChat3-4B.

You are a proactive visual assistant helping a user perform a procedural task. Use the current observation and previous guidance to determine whether new guidance is needed at the requested guidance level. If guidance is needed, output `</Response>` followed by one concise instruction that helps achieve the task goal. If no useful new guidance is needed, output `</Silence>`.

Based on the current observation and previous guidance, determine whether new guidance is needed at the requested guidance level. If needed, provide one concise instruction for the task.

MMDuet2.

You are a proactive assistant helping a user perform a procedural task. Use the current observation and previous guidance to determine whether new guidance is needed at the requested guidance level.

Based on the current observation and previous guidance, determine whether new guidance is needed at the requested guidance level. If needed, provide one short imperative instruction for the task.

Figure 23: Prompts for streaming and proactive assistance models.

Our Multi-Level Model

You are a proactive procedural assistant watching a first-person video stream. Use the goal, observed frames, and prior outputs to decide whether to guide. Do not repeat a guide or guide an already finished unit.

Output exactly `<silent>` when no new guidance is needed. Otherwise output `<response>` followed only by the guidance entries needed now, separated by “/”:

phase P: guidance / step P-S: guidance / action P-S-A: guidance

Positive numbered IDs express parent-child relationships. Number units in temporal order within the clip and restart child numbers under each parent. Put entries in Phase, Step, Action order and omit inactive levels.

Figure 24: Evaluation prompt for our model.

F DETAILS OF METRICS

We provide detailed descriptions of sF1, gF1, and PQS. sF1 considers both the semantic alignment between generated and GT guidance and the coverage of GT guidance through temporal-semantic matching (Liu et al., 2026). gF1 considers both response F1 and silent F1 at anchor points. PQS

reflects the correctness of response–silence decisions and the quality of generated guidance (Kundu et al., 2026). All metrics are computed independently for each dataset and guidance level.

F.1 SOFT F1 (SF1)

Following GuideMe (Liu et al., 2026), we partition each video at the midpoints $b_j = (\tau_j + \tau_{j+1})/2$ between consecutive GT response timestamps τ_j and τ_{j+1} . Within each segment, we perform minimum-cost one-to-one matching between predicted responses and GT anchors. For a predicted response $(\hat{y}_i, t_i)$ and a GT response (y_j, τ_j) , the semantic similarity s_{ij} and matching cost c_{ij} are

$$s_{ij} = |\cos(e(\hat{y}_i), e(y_j))|, \quad c_{ij} = (1 - s_{ij}) + [1 - \exp(-0.01(t_i - \tau_j)^2)]. \quad (2)$$

Here, e is the `all-mpnet-base-v2` sentence encoder, and timestamps are measured in seconds. Since Silent Anchors contain no GT guidance, we set the semantic term $(1 - s_{ij})$ to 1 when matching a predicted response to a Silent Anchor. The final matching set $\mathcal{M}$ includes only pairs in which both the prediction and GT are responses.

sPrecision (sP) measures how well the generated responses semantically match the GT responses, while sRecall (sR) measures how completely the GT responses are covered. Let P and R denote the numbers of predicted and GT responses, respectively, and C the sum of semantic similarities over matched pairs. We compute sPrecision and sRecall by dividing C by P and R , respectively, and sF1 as their harmonic mean:

$$C = \sum_{(i,j) \in \mathcal{M}} s_{ij}, \quad \text{sP} = \frac{C}{P}, \quad \text{sR} = \frac{C}{R}, \quad \text{sF1} = \frac{2 \text{sP} \cdot \text{sR}}{\text{sP} + \text{sR}} = \frac{2C}{P + R}. \quad (3)$$

We multiply sPrecision, sRecall, and sF1 by 100 when reporting results. Predicted responses without a matched GT response lower sPrecision, while GT responses without a matched prediction lower sRecall. The temporal cost affects the matching assignment but does not directly penalize the final sF1 score.

F.2 G-MEAN F1 (GF1)

gF1 is computed from response–silence decisions at the original anchors. At a GT Response Anchor, a predicted response counts as TP and predicted silence as FN. At a GT Silent Anchor, predicted silence counts as TN and a predicted response as FP. Response F1 (RF1) treats response as the positive class and evaluates whether the model responds when guidance is needed. Silent F1 (SF1) treats silence as the positive class and evaluates whether the model remains silent when guidance is not needed. Following PWR (Kundu et al., 2026), we compute gF1 as the geometric mean of these two scores:

$$\text{RF1} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}, \quad \text{SF1} = \frac{2\text{TN}}{2\text{TN} + \text{FP} + \text{FN}}, \quad \text{gF1} = 100 \sqrt{\text{RF1} \cdot \text{SF1}}. \quad (4)$$

gF1 does not use temporal matching or assess response content; an inaccurate response at a GT Response Anchor still counts as TP.

F.3 PROACTIVE ASSISTANCE QUALITY SCORE (PQS)

PQS jointly evaluates decision correctness and response quality at the original anchors. For each TP response, GPT-5.2 evaluates the generated guidance against the GT guidance using the evaluation prompt and scoring rubric from PWR (Kundu et al., 2026). The judge assigns ratings $r_{i,d} \in 1, \dots, 5$ for relevance, specificity, actionability, and conciseness. We normalize the mean rating to a content score $g_i \in [0, 1]$ and compute PQS as

$$\bar{r}_i = \frac{1}{4} \sum_{d=1}^4 r_{i,d}, \quad g_i = \frac{\bar{r}_i - 1}{4}, \quad \text{PQS} = \frac{100}{N} \left(\text{TN} + \sum_{i \in \mathcal{T}} g_i \right). \quad (5)$$

Here, $\mathcal{T}$ is the set of TP responses, and $N = \text{TP} + \text{TN} + \text{FP} + \text{FN}$ is the total number of anchors. Each anchor receives a score of 1 for TN (correct silence), g_i for a TP response, and 0 for FP or FN. The final score is the average over all anchors, multiplied by 100. On ProactiveCoachBench, which contains equal numbers of Response and Silent Anchors, an always-silent predictor obtains a PQS of 50 but a gF1 of 0. We therefore interpret PQS alongside gF1.

G ADDITIONAL QUANTITATIVE RESULTS

As discussed in Section F, our evaluation metrics are computed from sPrecision, sRecall, Response F1, and Silence F1. Tables 9–12 report these four metrics in detail. Table 9 shows that our method achieves a better balance between Response F1 and Silence F1, together with substantial gains in sPrecision. Tables 10–11 further show the effects of hierarchical supervision and different supervision levels. Table 12 provides the results for zero-shot and fine-tuned multi-level guidance.

Model	Ours	ProactiveCoachBench												EgoProactive			
		Phase				Step				Action				Step			
		sPrecision	sRecall	Response F1	Silence F1	sPrecision	sRecall	Response F1	Silence F1	sPrecision	sRecall	Response F1	Silence F1	sPrecision	sRecall	Response F1	Silence F1
Proprietary Frontier Models																	
Gemini 3 Flash Preview		53.05	39.68	66.25	74.73	41.47	44.11	59.66	58.35	39.92	54.68	65.26	50.59	52.05	33.04	60.74	68.91
Claude Haiku 4.5		35.41	60.87	64.90	11.82	32.05	62.78	65.77	7.73	32.12	62.96	66.14	8.64	35.13	46.02	63.84	43.39
GPT-5.4 mini		33.06	38.50	55.81	44.84	33.41	36.17	54.25	48.33	37.06	35.83	53.94	53.80	39.05	43.05	61.03	47.08
Guidance Assistant Models																	
ProAssist-8B		21.55	8.22	22.33	67.63	28.19	6.28	17.14	66.33	37.18	6.28	14.48	66.17	19.18	3.83	9.64	63.02
Streaming Models																	
VideoLLM-Online		1.06	0.30	1.21	66.72	2.16	0.18	0.63	66.66	4.95	0.18	0.54	66.68	2.17	0.30	1.41	63.27
MMDuet2		15.11	8.73	27.08	60.30	26.16	19.90	42.45	54.85	27.98	19.02	43.16	56.20	32.04	33.28	52.56	38.03
VideoChat3-4B		29.93	49.55	60.62	6.03	27.80	50.95	63.34	10.37	29.08	57.83	65.61	4.94	34.59	48.71	61.12	21.59
Multimodal Models																	
VideoChat-R1.5-7B		0.09	0.04	0.09	66.67	0.66	0.04	0.23	66.54	1.41	0.06	0.17	66.29	0.62	0.12	0.34	65.02
InternVL3-8B		19.25	12.23	34.11	59.66	31.73	24.23	48.41	56.15	37.34	37.08	57.48	52.42	39.95	12.63	23.89	60.71
VideoLLaMA3-7B		19.75	15.40	37.38	52.87	28.13	20.77	43.33	54.99	33.90	27.95	48.30	51.93	2.88	0.73	1.36	61.27
LLaVA-Video-7B		23.34	20.20	43.18	52.79	33.51	32.17	53.36	50.00	39.21	22.15	44.54	59.67	41.00	34.89	48.50	46.82
Qwen3-VL-32B		34.84	56.47	64.77	22.89	29.86	57.66	65.45	8.72	29.88	59.25	66.23	6.55	44.01	28.20	44.73	55.47
Qwen3-VL-4B		29.37	50.65	63.88	10.43	28.41	54.59	65.45	10.34	29.04	56.56	66.05	11.68	35.86	48.62	59.59	21.63
	✓	73.95	44.26	68.88	80.38	59.13	44.45	68.04	75.30	46.69	56.37	68.22	61.34	47.66	31.86	60.00	65.69
Qwen3-VL-8B		32.56	59.98	66.44	1.37	29.24	59.49	66.47	1.38	28.62	58.92	66.60	1.21	32.94	59.47	70.02	0.09
	✓	74.08	45.87	71.33	81.41	58.70	46.40	69.05	75.12	46.55	57.65	68.05	60.41	50.38	29.57	58.78	67.65
Qwen3.5-4B		23.00	18.98	42.70	53.64	28.08	19.43	41.67	57.86	34.81	18.77	38.16	59.91	34.34	10.35	21.77	61.80
	✓	72.89	44.05	69.86	80.83	59.73	44.23	69.02	75.68	47.18	57.04	68.56	61.15	48.56	30.44	56.32	64.04
Qwen3.5-9B		32.14	58.17	65.30	2.08	30.19	56.64	64.38	11.24	31.69	61.09	64.98	11.10	29.68	51.08	63.18	10.57
	✓	72.37	41.48	66.52	79.65	61.57	41.72	66.92	75.94	49.19	54.43	67.77	64.61	48.43	31.75	58.98	65.52

Table 9: Detailed metrics for the main results in Table 2.

Supervision	ProactiveCoachBench Step				EgoProactive			
	sPrecision	sRecall	Response F1	Silence F1	sPrecision	sRecall	Response F1	Silence F1
<i>Qwen3-VL-4B</i>								
Step-only	40.72	48.23	66.87	59.99	50.61	27.95	55.10	66.30
Multi-level (Ours)	59.13	44.45	68.04	75.30	47.66	31.86	60.00	65.69
<i>Qwen3-VL-8B</i>								
Step-only	48.58	46.09	66.02	66.22	50.18	28.67	56.88	67.09
Multi-level (Ours)	58.70	46.40	69.05	75.12	50.38	29.57	58.78	67.65
<i>Qwen3.5-4B</i>								
Step-only	42.76	41.77	63.85	65.22	47.85	30.92	57.02	64.42
Multi-level (Ours)	59.73	44.23	69.02	75.68	48.56	30.44	56.32	64.04
<i>Qwen3.5-9B</i>								
Step-only	44.82	42.89	63.77	66.69	51.17	26.64	53.07	66.69
Multi-level (Ours)	61.57	41.72	66.92	75.94	48.43	31.75	58.98	65.52

Table 10: Detailed metrics for hierarchical vs. step-only supervision in Table 3.

Supervision	ProactiveCoachBench Step				EgoProactive			
	sPrecision	sRecall	Response F1	Silence F1	sPrecision	sRecall	Response F1	Silence F1
step-only	42.76	41.77	63.85	65.22	47.85	30.92	57.02	64.42
phase-step	48.32	51.12	67.89	64.59	48.29	29.26	55.16	64.39
step-action	56.57	41.09	66.56	75.32	45.60	35.52	61.19	62.32
phase-step-action	59.73	44.23	69.02	75.68	48.56	30.44	56.32	64.04

Table 11: Detailed metrics for the supervision-level ablation in Table 4.

FT	Phase				Step				Action			
	sPrecision	sRecall	Response F1	Silence F1	sPrecision	sRecall	Response F1	Silence F1	sPrecision	sRecall	Response F1	Silence F1
	15.06	6.76	15.17	69.69	14.71	6.05	13.22	64.65	29.21	59.74	63.38	1.94
✓	72.89	44.05	69.86	80.83	59.73	44.23	69.02	75.68	47.18	57.04	68.56	61.15

Table 12: Detailed metrics for zero-shot multi-level guidance in Table 5.

H QUALITATIVE RESULTS

Figures 25–30 present six qualitative examples from ProactiveCoachBench and EgoProactive. Compared with the baselines, our model more consistently provides appropriate guidance at response anchors while remaining silent at silent anchors. Its responses are also generally well aligned with the ground-truth guidance.



Figure 25: Qualitative comparison on ProactiveCoachBench-1 (goal: make a peanut butter and jelly tortilla roll).



Figure 26: Qualitative comparison on ProactiveCoachBench-2 (goal: disassemble and clean a bicycle seat stay assembly).



Figure 27: Qualitative comparison on ProactiveCoachBench-3 (goal: remove dry sticks from a fence).



Figure 28: Qualitative comparison on ProactiveCoachBench-4 (goal: perform a COVID-19 rapid antigen test).

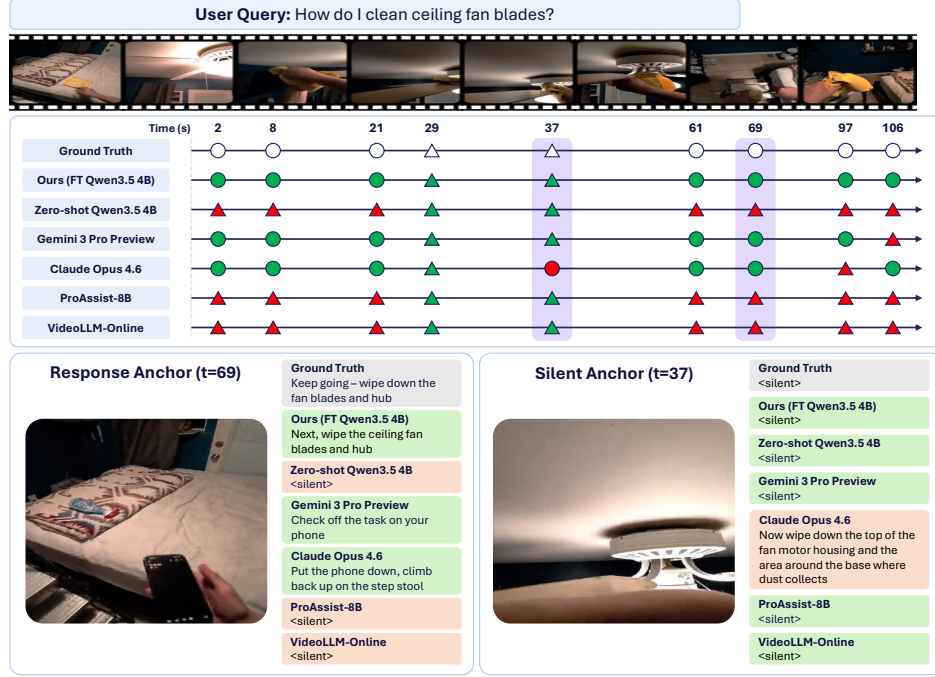


Figure 29: Qualitative comparison on EgoProactive-5 (goal: clean ceiling fan blades).

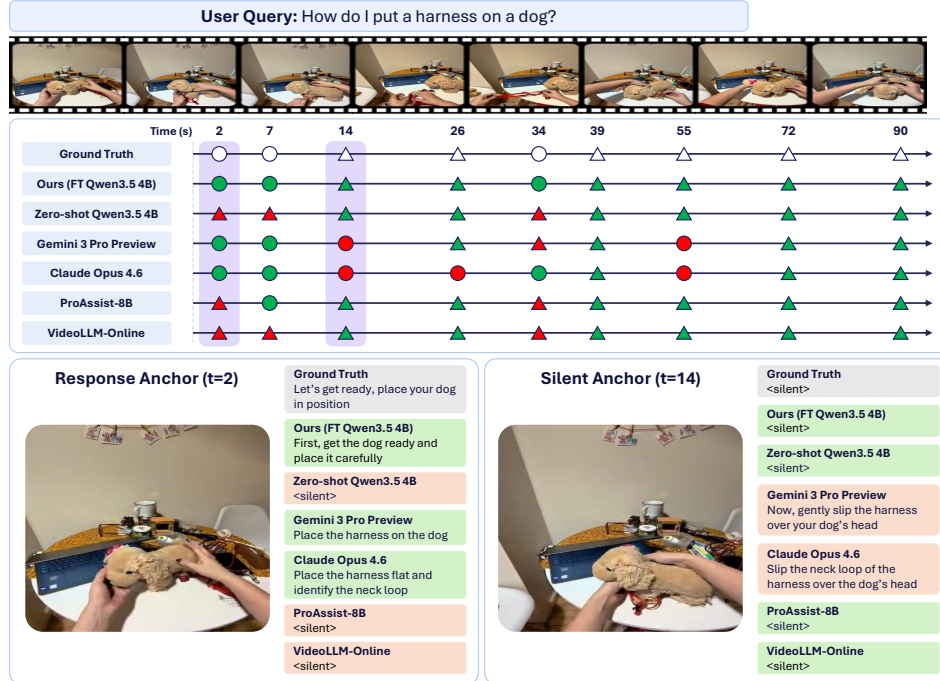


Figure 30: Qualitative comparison on EgoProactive-6 (goal: put a harness on a dog).

I LIMITATIONS

Our evaluation uses ground-truth dialogue history rather than model-generated outputs. When the model uses its own output history, further evaluation is needed to assess error accumulation and its ability to revise plans. In addition, guidance-level adaptation focuses on user-requested changes. Further research could explore how to infer the user’s expertise and understanding to adjust the guidance level automatically, potentially making assistance more useful in real-world interactions. Also, since some activities can be performed in different orders to achieve the same goal, evaluation methods that consider these alternatives are needed. Future work should extend training and evaluation to support long-term task tracking and plan revision, personalized guidance, and multiple valid execution orders.